

POSET ASSOCIAHEDRA AS REFINEMENTS OF HYPERGRAPHIC POLYTOPES

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ABSTRACT. We define a poset gradient to be a collection of acyclic oriented hypergraphs with vertex sets inside the antichains of a poset, and directed hyperedges determined by the poset structure.

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1. INTRODUCTION

The definition of poset associahedra uses tubes that are connected, filled, lower sets. Filled lower sets are those which for each boundary included in the tube, some element of the bundle of elements sharing that boundary must be included as well. Here we point out that a tubing on a poset always induces an acyclic orientation of the hypergraph of its elements with hyperedges the parts of bundles that are inclusions of immediate boundaries.

2. HYPERGRAPHIC POLYTOPES

This section is a summary of results in [BBM19] and [BP26].

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A hypergraph G on a set of vertices $V = V(G)$ is a subset of the power set of V . Usually the empty set and singletons are excluded, so the smallest allowed cardinality of the elements of G is 2. Alternatively the singletons are always required to be included. We will use the latter convention. The subsets that are listed for a given G are called the hyperedges of G .

If all the hyperedges are of cardinality 2, then G is a simple graph.

2.1. Orientations. An orientation $\mathcal{O}$ of a hypergraph G is a choice of an ordered bipartition of each hyperedge of cardinality greater than 2. A partial orientation leaves out some of the hyperedges, so is a choice of an ordered bipartition for some subset of the hyperedges. The first part of each ordered bipartition is called the head set of that hyperedge, or source, the second part is called the tail set, or target.

2.2. Acyclic Orientations. Requiring hyperedges to be subsets and orientations to be bipartitions means that there will be no overlap of head and tail, so no loops in the orientation. That's a good start but we need more to define acyclic orientations $\mathcal{O}$. To do so we choose the convention followed in [BBM19] to construct a new oriented pseudograph from any oriented hypergraph G . Starting with the set of nodes of G we collect as a new single node each equivalence class generated by $x \sim y$ if x and y are in a head set of some oriented hyperedge of G . The directed pseudograph $G/\mathcal{O}$ has nodes those equivalence classes and directed edges determined by the orientation of G : for each given hyperedge H of G we get new directed edge(s) from each node of $G/\mathcal{O}$ containing a head element of H to each node containing a tail element of H . Finally, the orientation $\mathcal{O}$ is acyclic if the directed pseudograph $G/\mathcal{O}$ has no cycles.

2.3. Flats and faces. A total orientation of a hypergraph is one for which each head set is a singleton. It turns out that the total orientations are the vertices of a convex polytope, the hypergraphic polytope P_G as studied in [BBM19]. For G with n nodes and c connected components P_G has dimension $n - c$. For connected hypergraphs the dimension is $n - 1$, and for multiple component hypergraphs the polytope is the cartesian product of the polytopes for each component. For G a simple graph P_G is the graphical zonotope, or parallelohedron (for the 5 examples in dimension 3: the permutohedron, the cube, the rhombic dodecahedron, the hexagonal prism, and the hexirhombic dodecahedron.)

Faces of the hypergraphic polytope P_G , in general, correspond to acyclic orientations of quotients-by-flats of the hypergraph G . A flat of G is a sub-hypergraph of G made up of any set of hyperedges of G which are each contained in a part of a partition of the nodes of G . That is, given the partition Π of the nodes of G we get a flat F which is a subset of the hyperedges of G ; those hyperedges which fall entirely in one of the parts of Π . The flat excludes any hyperedges that are not a subset of part of the partition. Then we can form the quotient G/F which contracts each hyperedge of the flat to a node. If the partition is just all the singletons of G then the quotient is G itself, all the hyperedges are preserved intact.

Faces of P_G correspond to acyclic orientations of G/F for flats of G . That is, we orient the hyperedges of G/F with any orientation $\mathcal{O}$ such that the resulting directed pseudograph $(G/F)/\mathcal{O}$ has no cycles.

Example. For example, consider the cycle graph $G = C_n$ with nodes $[n]$ in counting order clockwise. For $n = 4$ let the nodes be partitioned into $\Pi = \{\{1, 2\}, \{3, 4\}\}$. The flat F for this partition is the subgraph made of two components, both paths of 2 nodes. The quotient G/F is the two-node pseudo-hypergraph formed by collapsing these paths. There are two edges in the quotient, coming from the two edges of G not included in the flat F .

For $G = C_n$ the acyclic orientations of G/F for flats of G with exactly two components correspond to choosing any two edges of G and then choosing one direction, the same one, for both the edges in the quotient G/F . Those are often described as directed cuts of G . For $n = 4$ there are $\binom{4}{2}2 = 12$ of these; the 12 faces of the rhombic dodecahedron.

Indeed, the faces of P_G , as acyclic orientations of quotients-by-flats of a hypergraph G , correspond to certain partial orientations of G . However it can be hard to see which partial orientations can occur as faces without finding all the flats. This is a key part of our eventual result that tubings of a poset provide the faces of a refinement of the hypergraphic polytope.

2.4. Face poset. There are two quotients involved in finding a face of P_G , quotient by a flat (all of these are allowed) and then quotient by an orientation of that flat (to determine the acyclic orientations.) We have already pointed out that the vertices of P_G are the total acyclic orientations of G , which is the type for which both quotients are trivial: no hyperedges are collapsed and all head sets are singletons. At the other end the entire polytope (for a connected hypergraph, so dimension $n - 1$) is labeled by the un-oriented G , with the trivial flat from the trivial partition of one part.

In general the codimension of each face $(G/F, \mathcal{O})$, is the number of parts in the simplest partition giving that flat F plus the number of non-singleton heads in the orientation $\mathcal{O}$ of G/F , minus 1. Vertices have n parts for their flats and all heads singletons, so codimension $n - 1$ (dimension 0).

A face f of P_G is a subface of another face g when the label for f is an oriented quotient that refines the label for g . Refinement here is a series of steps in which either a) one non-singleton head set is split and a new tail chosen for one of the parts or b) a hyperedge is reconstituted (the flat is expanded) and given an orientation.

Or, we can preserve the heads and tails of the entire sets, even when considering orientations of flats. That is, replace the faces labeled by oriented flats with their partially oriented hypergraphs. Then we have for hyperedges $i \in H$, and two faces f_x and f_y corresponding to partial orientations x and y , that:

$$f_x \subset f_y \text{ when } \text{head}(y_i) \subseteq \text{head}(x_i), \text{ for all } x_i, y_i \text{ orientations of hyperedge } i.$$

2.4.1. Flip graph. In [BP26] the authors define flips on the total orientations, and show that the graph of these flips is the skeleton of the hypergraphic polytope.

2.5. Realization via Minkowski sums.

3. POSET ASSOCIAHEDRA

A finite poset $P, <$ will be represented by its Hasse diagram. A subposet of P is *connected* if it is connected in the Hasse diagram. A *lower* set of P is a subposet $L \subseteq P$ such that $y < x \in L \Rightarrow y \in L$.

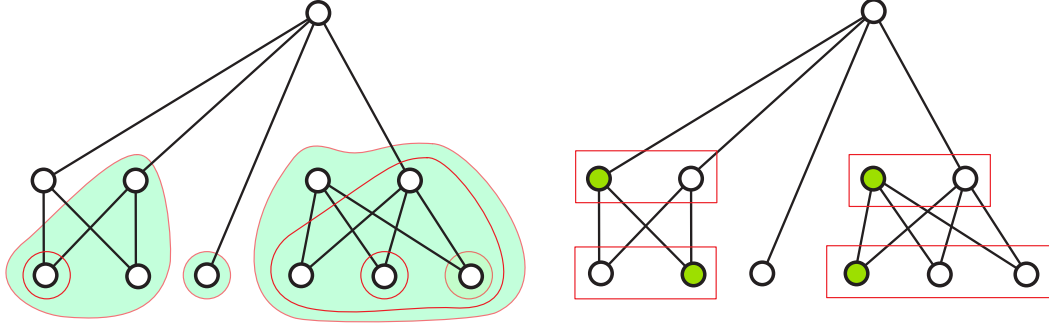


FIGURE 1. On the left is a maximal tubing T of a poset. On the right is the induced poset gradient, with the sets H_{xy} boxed (in red) and the elements $O_T(H_{xy})$ highlighted (in green).

From [DFRS15] recall that the *boundary* ∂x of an element $x \in P$ is the set of all elements that are strictly less than x . The *immediate boundary* of elements covered by x , which we denote $\partial_c x$, is the subset of the boundary directly covered by x in the Hasse diagram of P . That is, $\partial_c x$ is the set of maximal elements of ∂x . Of course $\partial_c x = \partial_c y$ iff $\partial x = \partial y$.

For any element x we define the *bundle* b_x as the collection of elements that have the same boundary as x . The *minimal bundle* is the set of minimal elements, which all have empty boundary. The bundles of P partition P , so if $y \in b_x$ then $b_x = b_y$.

From [1], a filled subposet F of P obeys the rule that if F contains the boundary of x then F intersects the bundle of x . That is, $\partial x \subseteq F \Rightarrow b_x \cap F \neq \emptyset$.

Definition 1. A *tube* t of a poset P is a connected, filled, lower set.

Definition 2. A *tubing* T of a poset P is a collection of tubes such that each pair of tubes is either nested or disjoint, and such that any union of tubes in T is filled.

3.1. Poset associahedra. The *poset associahedron* $\mathcal{KP}$ is the polytope whose faces are indexed by the tubings of P , ordered by reverse inclusion: vertices of $\mathcal{KP}$ correspond to maximal tubings of P . It is a simple polytope of dimension $|P| - b$ where b is the number of bundles in P .

3.2. Flip graph.

3.3. Summary of polytope families: big Venn.

4. THE MAIN THEOREM

5. BUNDLE ORIENTATION AND POSET GRADIENTS

NEED to work on this to expand it to the general faces.

Definition 3. Define a *total bundle orientation* $\mathcal{O}$ of P as follows: For each bundle b_x consider the sets $H_{xy} = \partial_c y$ for any distinct pair of elements $x, y \in P$ with $\partial_c y \subseteq b_x$. Also let $H_{xx} = b_x$ for ∂x non-empty. If $a \in b_x$ and $c \in b_y$ then $H_{xy} = H_{ac}$. A total bundle orientation $\mathcal{O}$ of P is a selection of a

single head element in each nonempty H_{xy} . That is, a function $\mathcal{O} : \{H_{xy} \mid x, y \in P\} \rightarrow P$ such that $\mathcal{O}(H_{xy}) \in H_{xy}$.

Definition 4.

This bit is true but unfinished: it appears that maybe all larger posets have gradients and gradient polytopes equivalent to the flattened version found by making a rank two poset with boundaries for each of the H_{xy} sets in the bigger poset. For a rank two graded poset P with all non-minimal bundles singletons, our definition of bundle orientation coincides with an orientation of a hypergraph in [BP26]. The nodes of the hypergraph are the minimal nodes of P and each maximal node y of P determines a hyperedge: $H_{xy} = \partial y \subseteq b_x$.

Definition 5. We define a *poset gradient* to be an acyclic bundle orientation $\mathcal{O}$ of a poset P , as follows: Let the Hasse diagram of P have edges directed upwards, towards the greater element. If new directed edges are added to the Hasse diagram so that each selected element $\mathcal{O}(H_{xy})$ is the source (head) of a directed edge to each of the other elements in H_{xy} , the resulting directed graph is acyclic.

Equivalently, each directed graph of just the new edges on just each bundle b_x is acyclic. Also equivalently, the function $\mathcal{O}$ obeys the classic definition of an acyclic directed hypergraph on each bundle.

Our main result is that each maximal tubing of P , which labels a vertex of the poset associahedron $\mathcal{KP}$, gives an acyclic bundle orientation, or gradient of P .

Definition 6. For a maximal tubing T on a poset P , the induced orientation $\mathcal{O}_T$ is defined:

$$\mathcal{O}_T(H_{xy}) = z \in H_{xy} \text{ s.t. } \forall t \in T, z \in t \Rightarrow H_{xy} \subseteq t$$

Theorem 7. For a maximal tubing T on a poset P , $\mathcal{O}_T$ defines a poset gradient.

Corollary 8. $\mathcal{KP}$ refines the dual fan of a polytope called the poset gradient polytope $\mathcal{GP}$.

Conjecture 9. The poset gradient polytope is actually a hypergraphic polytope, specifically the Cartesian product of the hypergraphic polytopes for each bundle, with hyperedges H_{xy} .

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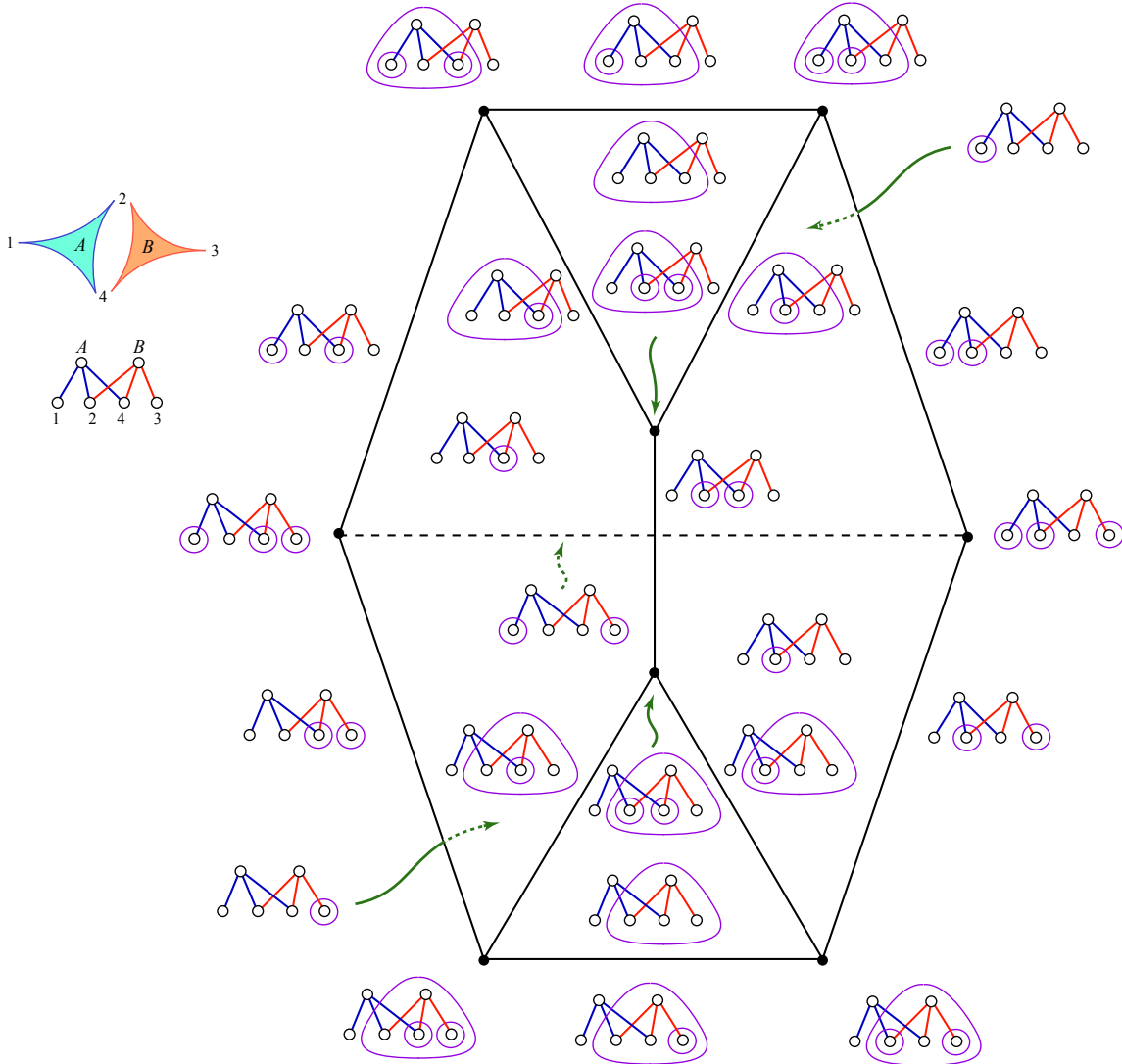


FIGURE 2. The poset associahedron for the poset with Hasse diagram shown. The corresponding hypergraph with two hyperedges A, B is shown, and in Figure 3 we show the poset gradient polytope, the hypergraphic polytope.

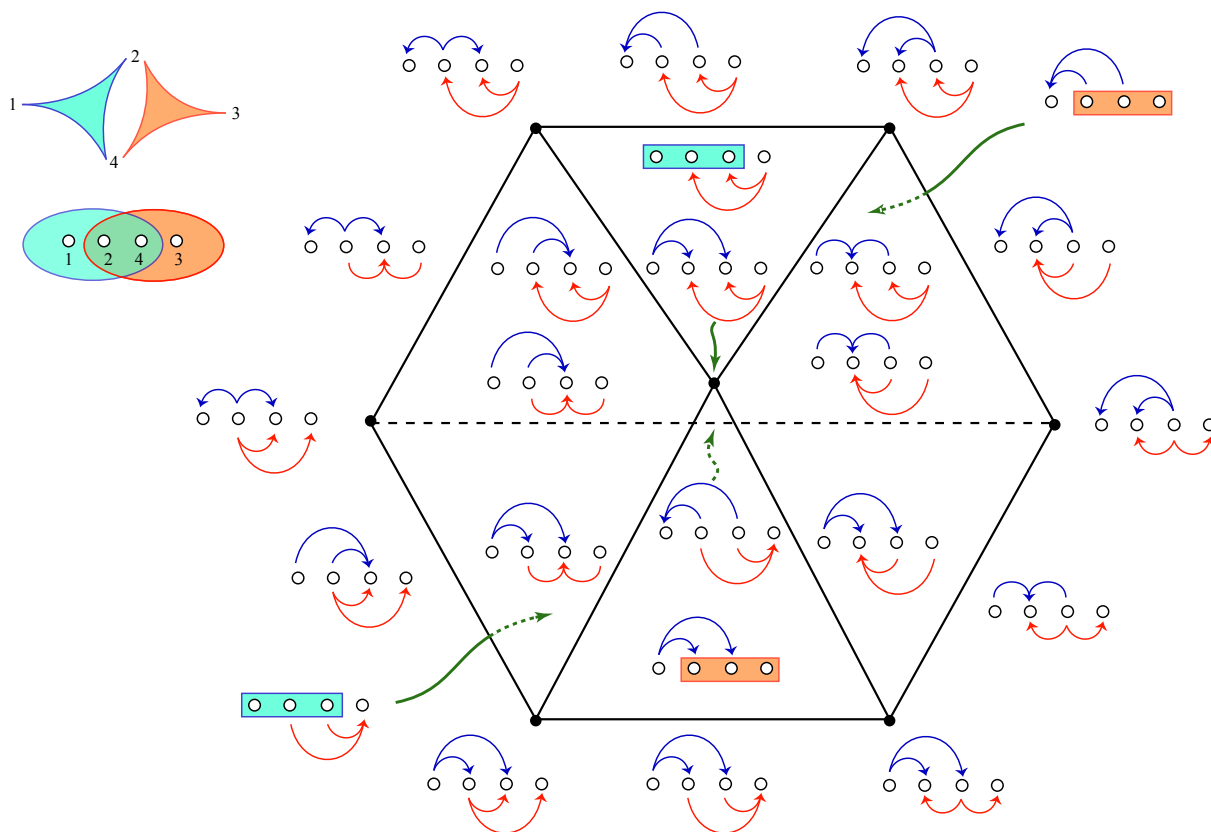


FIGURE 3. The hypergraphic polytope for the hypergraph shown. The central edge in the poset associatedhedron from Figure 2 has collapsed to a single vertex.

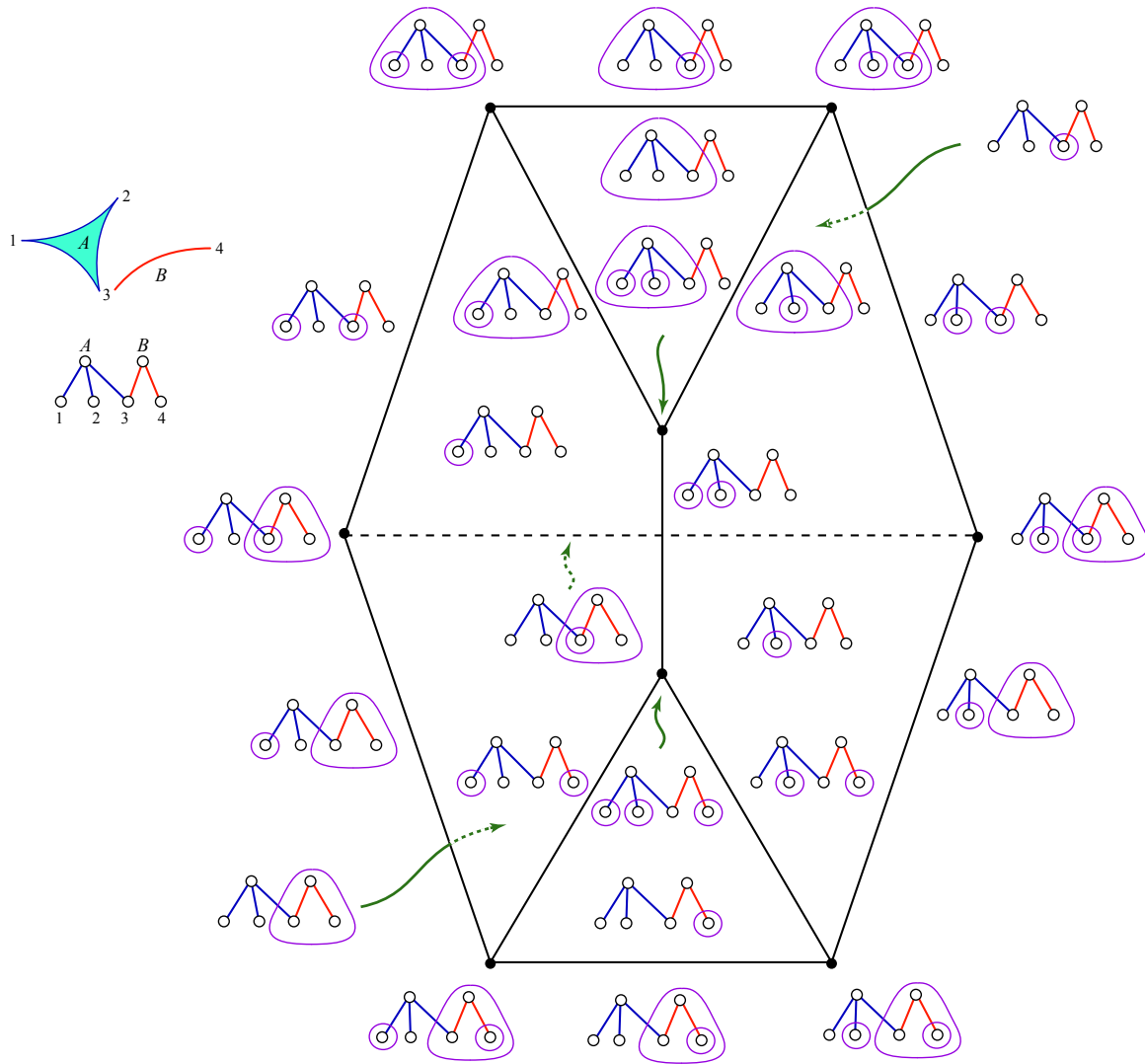


FIGURE 4. The poset associahedron for the poset with Hasse diagram shown. The corresponding hypergraph with two hyperedges A, B is shown, and in Figure 5 we show the poset gradient polytope, the hypergraphic polytope.

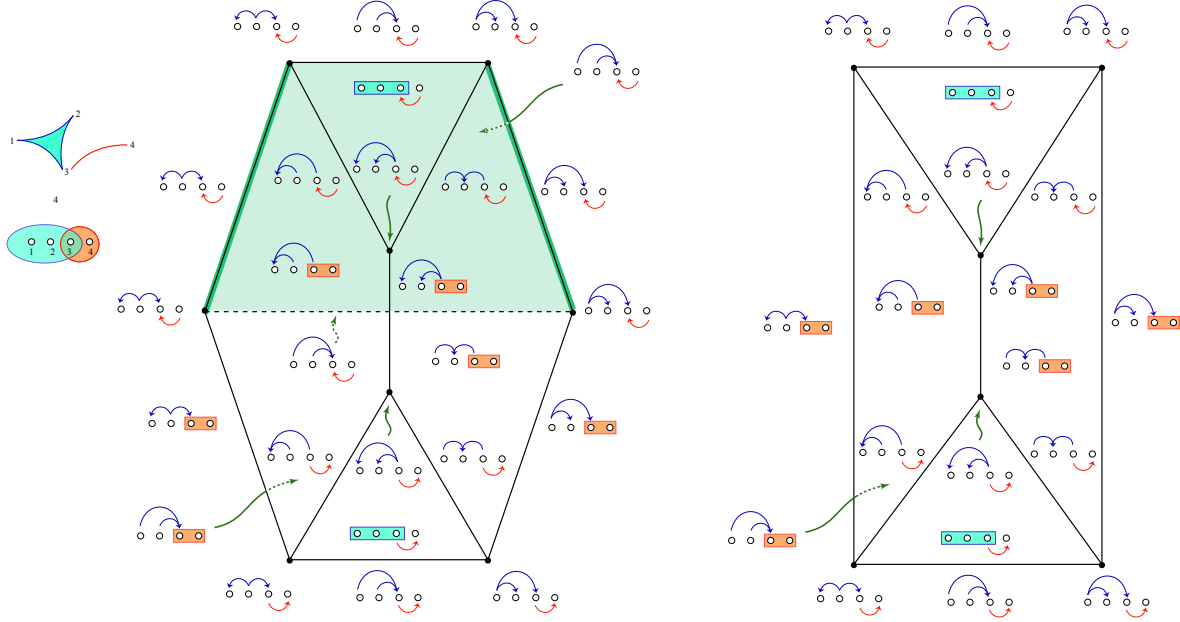


FIGURE 5. On the left is the poset associahedron from Figure 4, but labeled with the images of the gradient projection onto the hypergraph shown. On the right we have collapsed the faces that have identical oriented hypergraphs.

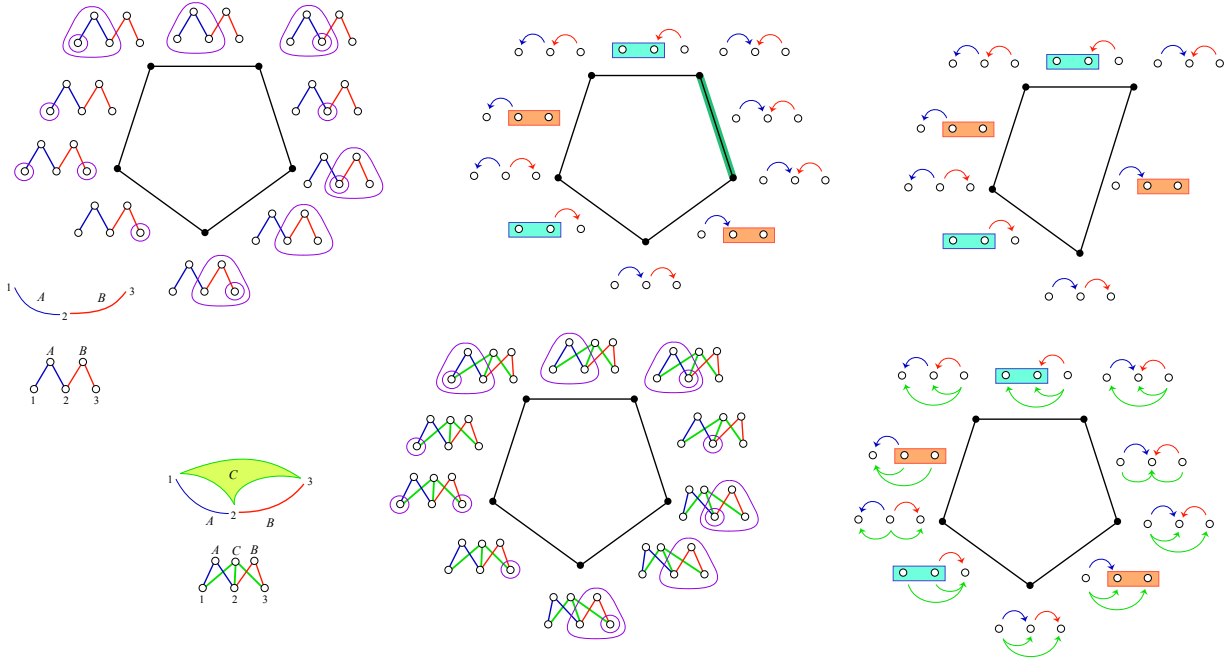


FIGURE 6. Two posets whose poset associahedron is the classic associahedron: the edges-and-nodes poset of the path graph and the same with an additional maximal node covering all the minimal nodes. These project onto two different hypergraphs, and so the resulting hypergraphic polytopes are respectively a graphic zonotope of the path graph (cubes) and again the associahedra, this time seen as interval hypergraphic polytopes. The general fact is that the poset with all intervals as its maximal nodes yields the interval hypergraphic polytope that is the classic associahedron.

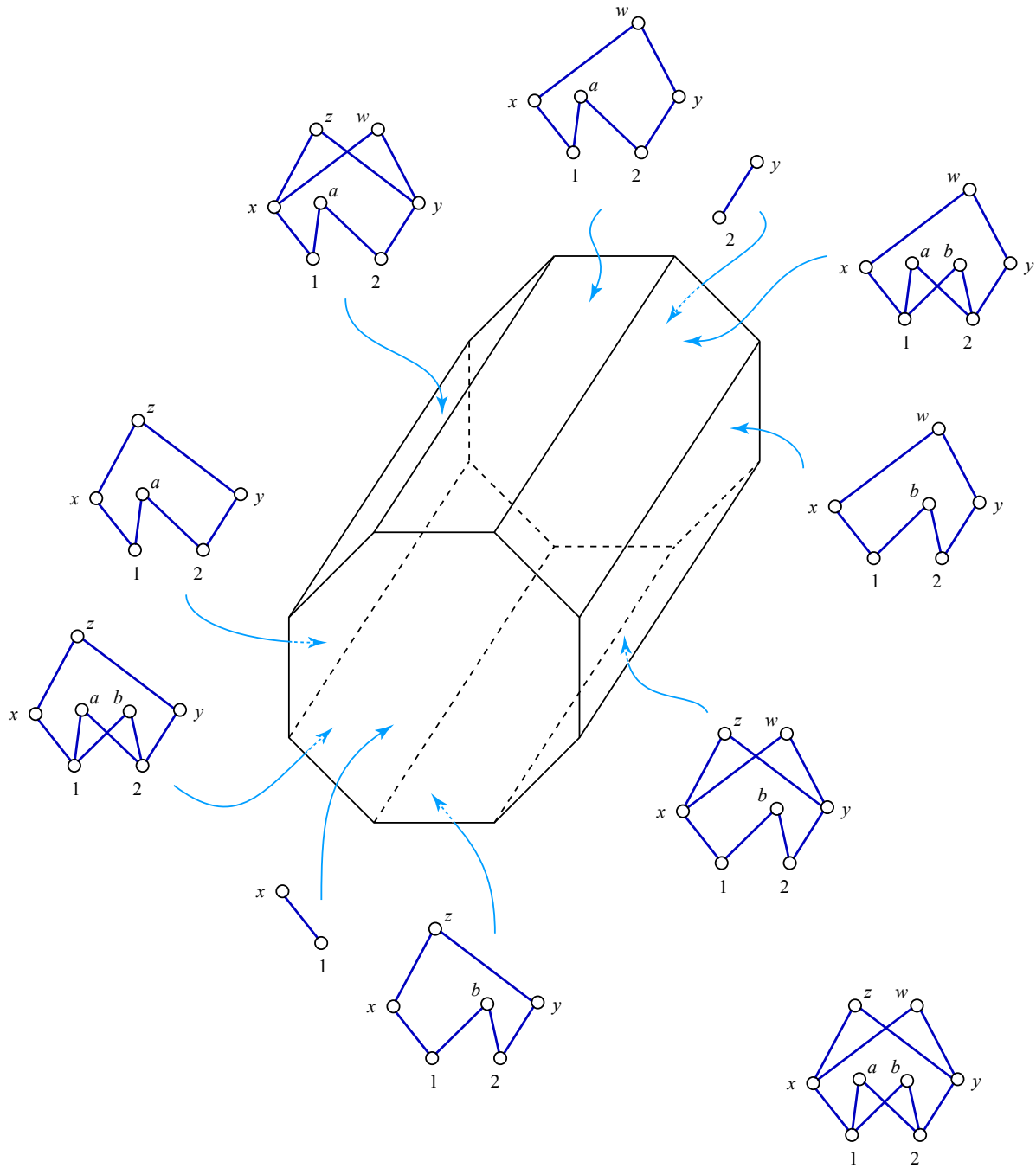


FIGURE 7. The poset associahedron for the poset with Hasse diagram shown at lower right. Tubes for the facets are shown as sub posets. More faces are shown in Figure 8.

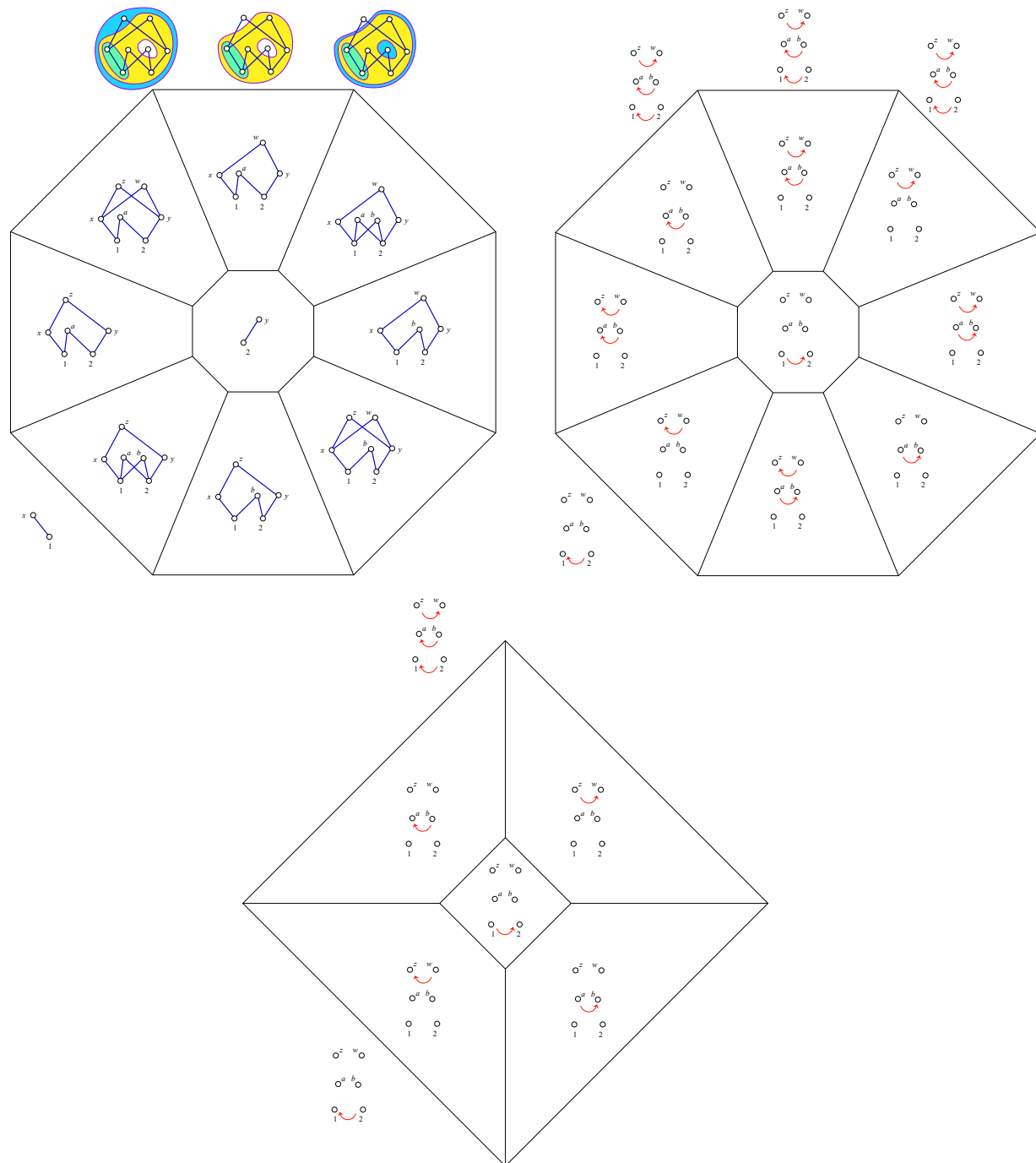


FIGURE 8. A flattened Schlegel diagram view of the 3D poset associahedron from Figure 7, the oriented hypergraphs, and the resulting hypergraphic polytope.